blinkX

BrainAThon NotifiMaster Challenge



Team NEXUS

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About BlinkX

BlinkX is a relatively new player in the Indian online stock broking market, launched in 2022. It's an initiative by JM Financial Services Ltd., a company with over 50 years of experience in the financial sector.

BlinkX's key features:

- Zero Brokerage: (Subscription Plans)
- Various Investment Options
- Mobile App(iOS and Android) and Website
- Learning Focus

Overall, BlinkX targets a growing segment of Indian investors looking for a user-friendly and affordable online trading platform with educational resources. They leverage the established reputation of JM Financial to provide a sense of security and trust.



Problem Statement

Objective:

- Develop a comprehensive push notification strategy to engage and retain users of an existing digital stock broking app.
- Create a strategy that leverages user behaviour and preferences for personalized, timely, and actionable push notifications.



You have to come up with a detailed plan covering the following:

- Conduct a thorough market analysis to uncover successful push notification strategies from other sectors.
- Proposals should include detailed strategies, expected outcomes, and a roadmap for implementation.
- Focus on how notifications will enhance user engagement and retention.

User Persona



New Entrant

Rahul, 21 year old student has low to moderate income level staying in Bangalore.

Goals: Learning about investing, building wealth over time, financial independence

Challenges Limited knowledge of investing, risk aversion, managing student loans and other debts

Motivations: Desire for financial security, influenced by social media and peer experiences, interested in ESG (Environmental, Social, Governance) investments



Mid-Income

Vijaya, 30 years old small business owner has an income of 80000 Rs Monthly lives in Agra ,U.P.

Goals: Saving for retirement, children's education, buying a home, building a safety net

Challenges: Balancing family expenses with saving and investing, managing existing debt, understanding complex financial products

Motivations: Desire for stable growth, financial security for family, tax efficiency



High-Income

Vikas , 52 years old Large business owner has an income of 500000 Rs Monthly lives in Bandra ,Mumbai.

Goals: Wealth preservation, estate planning, philanthropic goals, diversifying portfolio, enjoying a comfortable lifestyle

Challenges: Complex financial situations, managing large portfolios, tax optimization, legacy planning

Motivations: Maintaining and growing wealth, ensuring financial security for future generations, strategic philanthropy

Current Trends and User Behaviour



Real-Time Market Updates

Increasing reliance on real-time data for making informed trading decisions. High engagement with noifications delivering real-time price alerts, market news and trade opportunities.



Personalised Notifications

Personalized content leads to higher user engagement and satisfaction. Preference for notifications tailored to individual trading patterns, interests, and portfolio.



Humorous Notifications

Humorous content can enhance user engagement and make the app experience more enjoyable. Positive reception to witty and light-hearted notifications.



Time and Frequency Optimisation

Optimal timing and controlled frequency improve user engagement without causing notification fatigue. Preference for notifications sent at optimal times, avoiding overload.



Security Alerts

Heightened concern for security and regulatory compliance. Immediate attention to notifications regarding account security, suspicious activities, and compliance updates.

Case Studies

from other sectors

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zomato

Zomato sends contextual notifications based on user preferences and location, recommending nearby restaurants, new openings, and trending cuisines. It utilizes Hinglish notifications, blending Hindi and English, to make the experience enjoyable and relatable for users.

Key Takeway

Leveraging contextual and location-based data in notifications, along with Hinglish messaging, can provide relevant and enjoyable information, improving user experience and driving engagement.

amazon

Amazon sends personalized notifications based on user browsing history and purchase behavior. Notifications include tailored product recommendations, deals on viewed items, and reminders for saved items in the cart.

Key Takeway

Utilizing user data to deliver personalized recommendations and exclusive deals can enhance user satisfaction and drive conversions.

duolingo

Duolingo's language-learning app sends notifications to remind users to practice daily. Gamified notifications congratulate users on their progress and encourage them to maintain streaks.

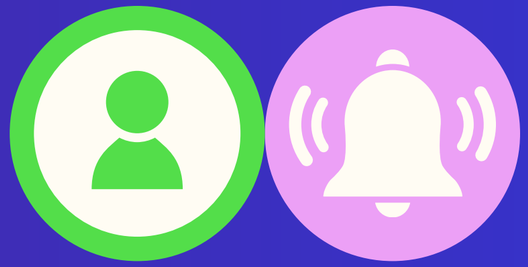
Key Takeway

- Incorporating gamification elements and progress tracking in notifications can encourage user engagement and habitual usage.

Pain Points and Solutions

Notification Fatigue: Users receive too many notifications, leading to annoyance and potential app uninstallation.	Implement **frequency capping** and allow users to customize their notification preferences. Use data analytics to determine optimal sending times to avoid overwhelming users.
Lack of Personalisation: Generic notifications fail to engage users effectively.	Use **AI** to tailor notifications based on user behaviour and preferences. **Segment users** and send targeted, relevant notifications.
Poor Timing: Notifications sent at inconvenient times may be ignored or dismissed.	Use **data-driven insights** to determine the best times to send notifications. Allow users to set their preferred notification times.
Irrelevant Content: Users receive notifications that are not relevant to their interests or trading activities.	Continuously **analyse user data** to ensure notifications are relevant and timely. Solicit **user feedback** to improve content relevance.

Notifications



Personalisation

Users Segregation for Notification

Based on Income	Based on Market & Time	Based on Location	Based on Data
• New Entrants • Middle Income Individuals • High Income Individual By considering past trading activities and trade value	• Shares Going Up or Down • 52 Weeks HIgh or Low • Market Open Time • Market Rush Hours • Market Closing Time	• Blend of Message with Hindi or other regional language based on location or user preference • Connecting the message with locations data for connect	• Based on user's watchlist and search • Based on users past financial achievements or failures like profits or loss in specific category of stocks

TYPES AND EXAMPLES OF NOTIFICATIONS

Welcome and Onboarding Messages:

"Welcome to BlinkX! Let's turn your trading game from 'Thoda Sa Adjust Karo' to 'Full Paisa Vasool'! 💸"

Educational Content:

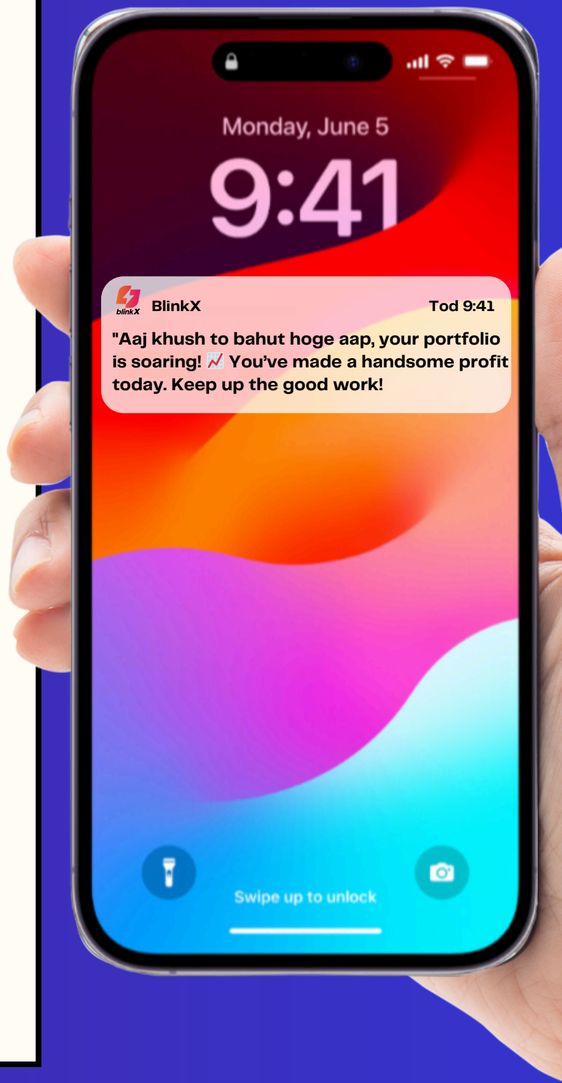
"Pro Tip: Keep an eye on market trends. ‘Ja Simran Ja, Market Trends Ke Peeche Bhaag!’ 📉🏃”

Market News:

""Market News: Tech stocks are booming today. ‘Mogambo khush hua!’ 📊😎"

Portfolio Performance:

"Aaj khush to bahut hoge aap, your portfolio is soaring! 📈 You’ve made a handsome profit today. Keep up the good work!



Technology Stack

Machine Learning (ML) & Natural Language Processing (NLP)

Data Collection

User behaviour: clicks, investments, watchlists.
Market trends, popular memes/trends/songs/quotes
Using tools like Google Analytics, Yahoo Finance API or custom scripts and a creative team which will analyse the data.

Model Training

Train an NLP model to analyse user data and market trends.
Python libraries such as TensorFlow, PyTorch, scikit-learn, and NLP libraries like spaCy or Hugging Face Transformers.
Fine-tune pre-trained NLP models (e.g., BERT, GPT) for our task.

Notification Generation

Select relevant message templates based on user segments and market conditions
Use data location to customize messages with relevant cultural references (memes, dialogues, song lyrics).
And make a system to validate these notifications

Implications



Data Collection

- **Integrate user activity tracking and market data feeds.**
- **Train NLP model with historical data and message templates.**
- **Continuously update model with new trends and user behavior.**



Notification Personalisation

- Develop user profiles (investment levels, locations, preferences).
- Create a library of message templates by user segment and market conditions.
- Integrate NLP model for real time content generation.



A/B Testing & Refinement

- Conduct A/B tests on different notification styles and content.
- Analyse user engagement metrics to find the most effective approach.
- Continuously refine NLP model and templates based on results.



Content Management

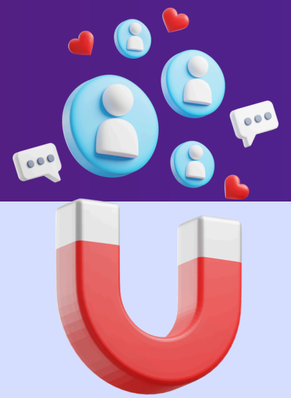
- **Curate and update memes, movie references, cultural content and market related information and news.**
- **Align content with user segments, current trends, and brand voice.**

SWOT

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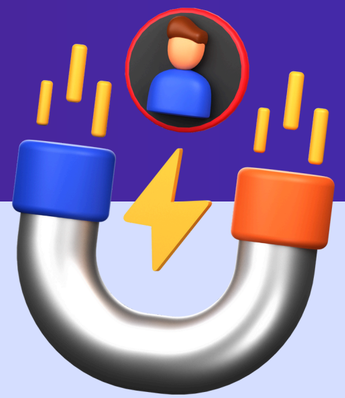
Strengths	Weaknesses	Opportunities	Threats
• **Personalisation** of users will give relevant notifications • **Timelines** which will help users in getting market insights at a glance. • **Relevance** from Humour, Cultural and Regional • **Actionable Insights** and Information • **AI** powered real time Notification Generation	• Reliance on user data and behaviour • **Technical Challenges** related to model • **Creative Fatigue** as maintaining the fresh trends and avoid repetitiveness can be challenging	• **Increase User Engagement** • **More retention** of user and increase in average time spent by user • **Driving Revenue Growth** • Enhanced **Brand Image** • **Data Driven Marketing** as user data can help in broader marketing strategies.	• **User Annoyance** if they don't like messages • **Competition** may adopt the similar techniques • **Technical Issues** while generation of notification or feeding up data to model • Personalisation may raise **privacy concerns or regulatory challenges**

Succes Metrics



Engagement

Click Through Rate(CTR) of Notifications
Session Duration after opening through Notifications



Retention

User Retention Rate,
Frequency of Use,
Churn Rate
during AB Testing



Investment Activity

Number of Trades,
Average Trade Value
Total Investment Growth

*All metrics are measured for a month

Release Plan



01 Preparation

- Data Collection
- Model Training
- Model Testing
- Content Creation
- Sample Notifications

02 Initial Testing

- Limited User Group (Beta Testing)
- Track Engagement Metrics
- Refining and Optimisation
- Customer Feedbacks

03 Phased Rollout

- Gradual release in all segment of users
- Focus on Success Metrics
- Communication & Education
- Growth in new users or opt-in

04 Improvements

- Regularly Monitoring
- Customer Behoviour and Feedback analysis
- Data-Driven Optimization
- AB Testing Iteration

Conclusion

- **Personalized & Timely Notifications**: Segmenting users and delivering relevant notifications based on investment level, location, market conditions, and preferences creates a more engaging user experience.
- **Humour & Cultural Relevance**: Leveraging memes, movie references, and trending content tailored to user location creates a more relatable and memorable brand image.
- **Actionable Insights**: Providing investment opportunities or educational resources within notifications empowers users and encourages investment activity within the app.
- **AI-powered Content Generation**: Utilizing **ML** and **NLP** allows for dynamic and creative notifications that adapt to market trends, keeping users informed and engaged.

In conclusion, this push notification strategy is not just about sending messages; it's about building a connection with our users. By delivering personalized, timely, and actionable information in a way that's both informative and entertaining, we can empower our users to make informed investment decisions and unlock the full potential of our digital stockbroking app. We are confident that this innovative approach will solidify our position as a leader in the market and drive long-term success for our company.